

Tanish Singh Rajpal

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Research interests: Reinforcement learning and algorithm design for quantum information systems, with an emphasis on entanglement routing in noisy, heterogeneous networks.

EDUCATION

Carnegie Mellon University

M.S. in Information Technology - Information Security
Information Networking Institute

Pittsburgh, PA

Expected May 2027

NMIMS University

B.Tech. in Computer Engineering
Mukesh Patel School of Technology Management & Engineering

Mumbai, India

Aug 2021 - May 2025

GPA: 3.83/4.00 | Valedictorian

RESEARCH EXPERIENCE

Reinforcement Learning for Heterogeneous Quantum Network Routing

Independent Study, Carnegie Mellon University

May 2026 - Present

Advisor: Prof. Michael Mattarock | Information Networking Institute

- Designed a decentralized Proximal Policy Optimization (PPO) routing agent in NetSquid with fixed observation and action dimensions across network sizes, using noisy local entanglement-fidelity signals.
- Benchmarked against centralized negative-log-fidelity Dijkstra and hop-count routing across homogeneous and heterogeneous quantum-hardware noise models.
- Across three training seeds, obtained $\geq 99\%$ mean entanglement fidelity on homogeneous grids from 3×3 to 10×10 and 90-93% under heterogeneous noise.
- Routing success was 100% in five of six configurations; the 10×10 homogeneous grid yielded $84.4\% \pm 27.0\%$ across seeds.
- Used stochastic evaluation to probe robustness beyond deterministic policy evaluation.

Manuscript in preparation.

A Hybrid Architecture for Post-Quantum EAP-TLS

Master's Thesis, Carnegie Mellon University

Aug 2026 - Present

Advisors: Prof. Pedro Bustamante and Prof. Michael Mattarock; Reader: Prof. Patrick Tague

- Investigating authentication failures associated with large ML-DSA certificates in WPA-Enterprise 802.1X deployments, including potential Path MTU Discovery blackholes.
- Designing and simulating an ns-3 architecture that combines edge QUIC termination with a QKD-secured backbone.
- Defined four evaluation hypotheses covering handshake latency, RADIUS compute offload, IPv4/IPv6 effects, and QKD key-delivery capacity during authentication bursts.

Evaluation in progress; defense and final report expected Spring 2027.

SELECTED QUANTUM COMPUTING PROJECT

Quantum Portfolio Optimization Using QAOA

Team Lead, Yale YQuantum Hackathon | Third place

Apr 2026

- Led problem formulation and circuit construction, mapping Markowitz portfolio optimization to a QUBO and then an Ising Hamiltonian for the Quantum Approximate Optimization Algorithm (QAOA), applied to insurance investment portfolios.

[Project repository](#)

PEER REVIEWED PUBLICATIONS

Journal article

Rajpal, T. S., and Naithani, A. (2026). NeuroCrypt: A Neuro Symbolic AI Ecosystem for Advanced Cryptographic Data Security and Transmission. *IEEE Transactions on Artificial Intelligence*, 7(1), 512-521. doi:10.1109/TAI.2025.3577605

- Developed a neuro-symbolic framework integrating a rule-based cryptographic engine and a deep neural network for dynamic symmetric-key generation.

Conference papers

Naithani, A., and **Rajpal, T. S.** (2025). Blockchain Based Digital Health Passport. *2025 IEEE Zooming Innovation in Consumer Technologies Conference (ZINC)*, 165-170. doi:10.1109/ZINC65316.2025.11100281

Naithani, A., **Rajpal, T. S.**, and Mistry, V. (2024). TAV-Cart: Transaction Assistance and Validation Cart System for Effortless Shopping and Checkout. *2024 International Conference on Advances in Computing, Communication and Materials (ICACCM)*, 1-6. doi:10.1109/ICACCM61117.2024.11059075

Naithani, A., Jain, V., **Rajpal, T. S.**, and Mistry, V. (2024). AI based Goalkeeper for Penalty Shot Prediction using RNN and Pattern Matching Algorithm. *2024 12th International Symposium on Digital Forensics and Security (ISDFS)*, 1-7. doi:10.1109/ISDFS60797.2024.10527307

RESEARCH METHODS AND TOOLS

Methods: PPO, reinforcement learning, network simulation, QAOA, QUBO and Ising formulations

Platforms: NetSquid, ns-3, Qiskit, Git and GitHub

Security and networking: EAP-TLS, ML-DSA, 802.1X, PMTUD, QKD, Burp Suite, Metasploit

INDUSTRY EXPERIENCE

Newton's Apple

Pune, India

Vulnerability Assessment and Penetration Testing Intern

Oct 2023 - Jan 2024

- Identified and supported remediation of three critical vulnerabilities in a confidential oil-company Android application using Burp Suite and Metasploit.
- Conducted static Java code analysis and worked with developers to remediate recurring security weaknesses in subsequent releases.

HONORS AND ADDITIONAL TRAINING

Quantum Excellence, IBM Qiskit Global Summer School

2026

Completed the full lab series, including the Advanced Track. [Credential](#)

First Place, Crypto Quest Cybersecurity Competition

2024

Tech-Fest Ambiora

ACADEMIC SERVICE

Student Ambassador, Information Networking Institute

Oct 2025 - Present

Carnegie Mellon University

Serve as a contact for prospective students and contribute to community-event planning.